APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Trigonometric Heights & Distances Case Study

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 10 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 93 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Solve real-world angle of elevation & depression problems including maritime lighthouse and dual-building case studies.

1. Executive Overview & Problem Context

Trigonometry ("triangle measurement") establishes exact mathematical ratios between interior acute angles and side lengths in right-angled Euclidean triangles. Applications in heights and distances enable geodetic surveyors and navigators to determine inaccessible vertical elevations and horizontal expanses using theodolites.

2. Core Analysis, Solutions & Systematic Working

- **Angles of Elevation vs Angles of Depression:**

  The Line of Sight connects an observer's ocular lens to the target object. When gazing upward above the horizontal datum, the acute angle formed is the Angle of Elevation (;,'à When looking downward below the horizontal datum, it is the Angle of Depression (<b'à By alternate interior angle geometry between parallel horizontals, Angle of Depression = Angle of Elevation.

- **Case Study 1: Coastal Lighthouse & Approaching Ships:**

  A lighthouse keeper 75m above sea level observes two ships aligned along the same line of sight. The angles of depression shift from 30° to 45°. Let d be the distance between ships. Tan(45°) = 75 / x -> x = 75m. Tan(30°) = 75 / (x + d) -> 1/" 3 = 75 / (75 + d) -> 75 + d = 75" 3 -> d = 75(" 3 - 1) "H 75(1.732 - 1) = 54.9 meters.

- **Case Study 2: Multi-Storey Building & Communication Tower:**

  From the top of a 7m building, the angle of elevation of a cable tower is 60° and the angle of depression of its base is 45°. Tan(45°) = 7/w -> horizontal distance w = 7m. Tan(60°) = h_tower/w -> " 3 = h_tower/7 -> h_tower = 7" 3 m. Total Tower Height = 7 + 7" 3 = 7(1 + " 3) "H 19.12 meters.

3. Detailed Data Matrix & Systematic Findings

Observational Scenario	Observer Height	Key Angle(s)	Trig Ratio Applied	Computed Physical Metric
Coastal Lighthouse	75 meters	30° and 45°	tan ;, Ò ÷ ÷6—FR ò F¦ 6Vç@	Ship separation = 54.9 meters
Telecom Tower on Hill	7.0 meters	60° Elev, 45° Dep	tan 60° = " 3, tan 45° = 1	Tower elevation = 19.12 meters
Hot Air Balloon Shadow	Ground theodolite	45° to 60°	sin ;, b F â ¸	Altitude = 120.5 meters
River Width Survey	Riverbank cliff (30m)	30° to opposite bank	tan 30° = 1/" 3	River width = 51.96 meters
Kite String Tension	Ground anchor	60° elevation	sin 60° = " 3 / 2	String length = 115.47 meters

4. Student Conclusions & Reflective Synthesis

Trigonometric surveying demonstrates how abstract trigonometric ratios translate directly into vital civil engineering, aeronautical flight path modeling, and satellite navigation benchmarks.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding trigonometric precision! Geometric ray diagrams, tangent ratio applications, and radical simplifications are excellent."

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